

Spring Data JPA with Spring Boot, Hibernate

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Spring Data JPA - Quick Example:

Application.properties:

```
application.properties × Country.java CountryRepository.java CountryService.java
1 # Spring Framework and application log
2 logging.level.org.springframework=info
3 logging.level.com.cognizant=debug
4
5 # Hibernate logs
6 logging.level.org.hibernate.SQL=trace
7 logging.level.org.hibernate.orm.jdbc.bind=trace
8
9 # Log pattern
10 logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %Sp %-25.25logger{25} %25M %4L %m%n
11
12 # Database configuration
13 spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
14 spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
15 spring.datasource.username=root
16 spring.datasource.password=root
17
18 # Hibernate configuration
19 spring.jpa.hibernate.ddl-auto=validate
20 spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
```

Country.java:

```
application.properties Country.java × CountryRepository.java CountryService.java
1 package com.cognizant.ormlearn.model;
2
3 import jakarta.persistence.Column;
4 import jakarta.persistence.Entity;
5 import jakarta.persistence.Id;
6 import jakarta.persistence.Table;
7
8 @Entity
9 @Table(name="country")
10 public class Country {
11
12     @Id
13     @Column(name="co_code")
14     private String code;
15
16     @Column(name="co_name")
17     private String name;
18
19     public String getCode() { return code; }
20     public void setCode(String code) { this.code = code; }
21     public String getName() { return name; }
22     public void setName(String name) { this.name = name; }
23
24     @Override
25     public String toString() {
26         return "Country [code=" + code + ", name=" + name + "];"
27     }
28 }
```

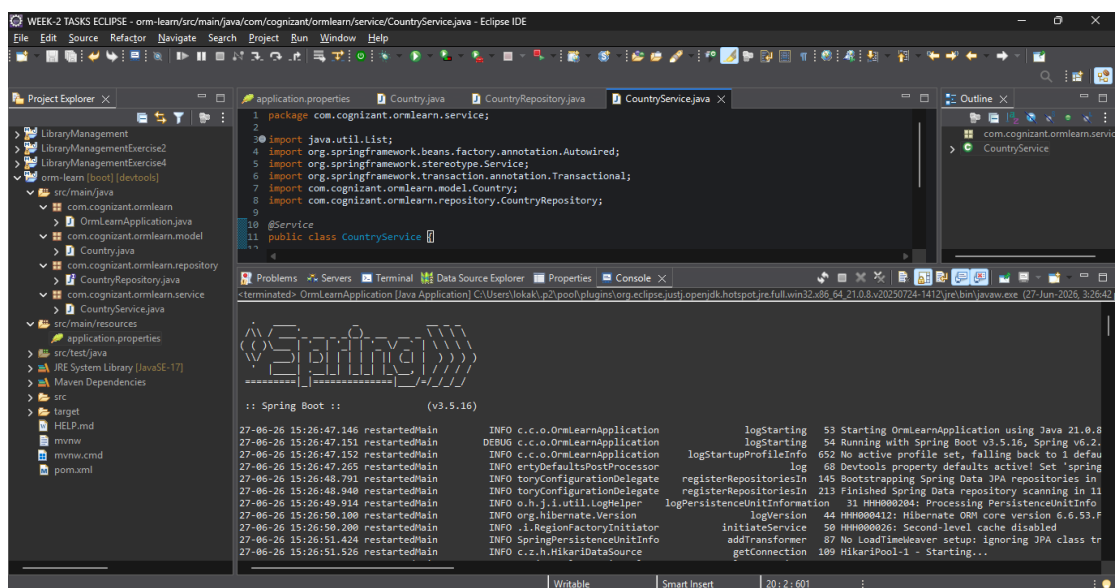
CountryRepository.java:

```
application.properties Country.java CountryRepository.java CountryService.java
1 package com.cognizant.ormlearn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5 import com.cognizant.ormlearn.model.Country;
6
7 @Repository
8 public interface CountryRepository extends JpaRepository<Country, String> {}
9 }
```

CountryService.java:

```
application.properties Country.java CountryRepository.java CountryService.java
1 package com.cognizant.ormlearn.service;
2
3 import java.util.List;
4 import org.springframework.beans.factory.annotation.Autowired;
5 import org.springframework.stereotype.Service;
6 import org.springframework.transaction.annotation.Transactional;
7 import com.cognizant.ormlearn.model.Country;
8 import com.cognizant.ormlearn.repository.CountryRepository;
9
10 @Service
11 public class CountryService {}
12
13 @Autowired
14 private CountryRepository countryRepository;
15
16 @Transactional(readOnly = true)
17 public List<Country> getAllCountries() {
18     return countryRepository.findAll();
19 }
20 }
```

Output:



The screenshot shows the Eclipse IDE interface. The Project Explorer on the left displays the project structure, including the 'src/main/java' directory with the following packages and classes:

- com.cognizant.ormlearn
- com.cognizant.ormlearn.application
- com.cognizant.ormlearn.model
- com.cognizant.ormlearn.repository
- com.cognizant.ormlearn.service
- com.cognizant.ormlearn.test

The Console window at the bottom shows the application's startup logs, including the Spring Boot version (v3.5.16) and the application's name (OmLearnApplication). The logs indicate that the application is running successfully and that the database is initialized.

Difference between JPA, Hibernate and Spring Data JPA

OrmlearnApplication.java:

```
1 package com.cognizant.ormlearn;
2
3 import java.util.List;
4 import org.slf4j.Logger;
5 import org.slf4j.LoggerFactory;
6 import org.springframework.boot.SpringApplication;
7 import org.springframework.boot.autoconfigure.SpringBootApplication;
8 import org.springframework.context.ApplicationContext;
9 import com.cognizant.ormlearn.model.Country;
10 import com.cognizant.ormlearn.model.Employee;
11 import com.cognizant.ormlearn.service.CountryService;
12 import com.cognizant.ormlearn.service.EmployeeService;
13
14 @SpringBootApplication
15 public class OrmLearnApplication {
16
17     private static final Logger LOGGER = LoggerFactory.getLogger(OrmLearnApplication.class);
18     private static CountryService countryService;
19     private static EmployeeService employeeService;
20
21     public static void main(String[] args) {
22         ApplicationContext context = SpringApplication.run(OrmLearnApplication.class, args);
23
24         countryService = context.getBean(CountryService.class);
25         employeeService = context.getBean(EmployeeService.class);
26
27         LOGGER.info("Inside main");
28
29         // Test existing Country retrieval
30         testGetAllCountries();
31
32         // Test Hands-on 4 Employee Creation
33         testAddEmployee();
34     }
35 }
```

```
20
21 public static void main(String[] args) {
22     ApplicationContext context = SpringApplication.run(OrmLearnApplication.class, args);
23
24     countryService = context.getBean(CountryService.class);
25     employeeService = context.getBean(EmployeeService.class);
26
27     LOGGER.info("Inside main");
28
29     // Test existing Country retrieval
30     testGetAllCountries();
31
32     // Test Hands-on 4 Employee Creation
33     testAddEmployee();
34 }
35
36 private static void testGetAllCountries() {
37     LOGGER.info("Start Country Test");
38     List<Country> countries = countryService.getAllCountries();
39     LOGGER.debug("countries={}", countries);
40     LOGGER.info("End Country Test");
41 }
42
43 private static void testAddEmployee() {
44     LOGGER.info("Start Employee Test");
45
46     Employee emp = new Employee("John Doe");
47     employeeService.addEmployee(emp);
48
49     LOGGER.debug("Saved Employee: {}", emp);
50     LOGGER.info("End Employee Test");
51 }
52 }
```

CountryService.java:

```
CountryServi... X Employee.java EmployeeRep... EmployeeS
1 package com.cognizant.ormlearn.service;
2
3 import java.util.List;
4
5
6
7
8
9
10 @Service
11 public class CountryService {
12
13     @Autowired
14     private CountryRepository countryRepository;
15
16     @Transactional(readOnly = true)
17     public List<Country> getAllCountries() {
18         return countryRepository.findAll();
19     }
20 }
```

Employee.java:

```
CountryServi... X *Employee.java X EmployeeRep... EmployeeServ... OrmLearnAppl... »
1 package com.cognizant.ormlearn.model;
2 import jakarta.persistence.Column;
3 import jakarta.persistence.Entity;
4 import jakarta.persistence.GeneratedValue;
5 import jakarta.persistence.GenerationType;
6 import jakarta.persistence.Id;
7 import jakarta.persistence.Table;
8
9 @Entity
10 @Table(name="employee")
11 public class Employee {
12     @Id
13     @GeneratedValue(strategy = GenerationType.IDENTITY)
14     @Column(name="id")
15     private Integer id;
16
17     @Column(name="name")
18     private String name;
19
20     // Default Constructor
21     public Employee() {}
22
23     // Constructor with parameters
24     public Employee(String name) {
25         this.name = name;
26     }
27
28     // Getters and Setters
29     public Integer getId() { return id; }
30     public void setId(Integer id) { this.id = id; }
31     public String getName() { return name; }
32     public void setName(String name) { this.name = name; }
33
34     @Override
35     public String toString() {
36         return "Employee [id=" + id + ", name=" + name + "];"
37     }
38 }
```

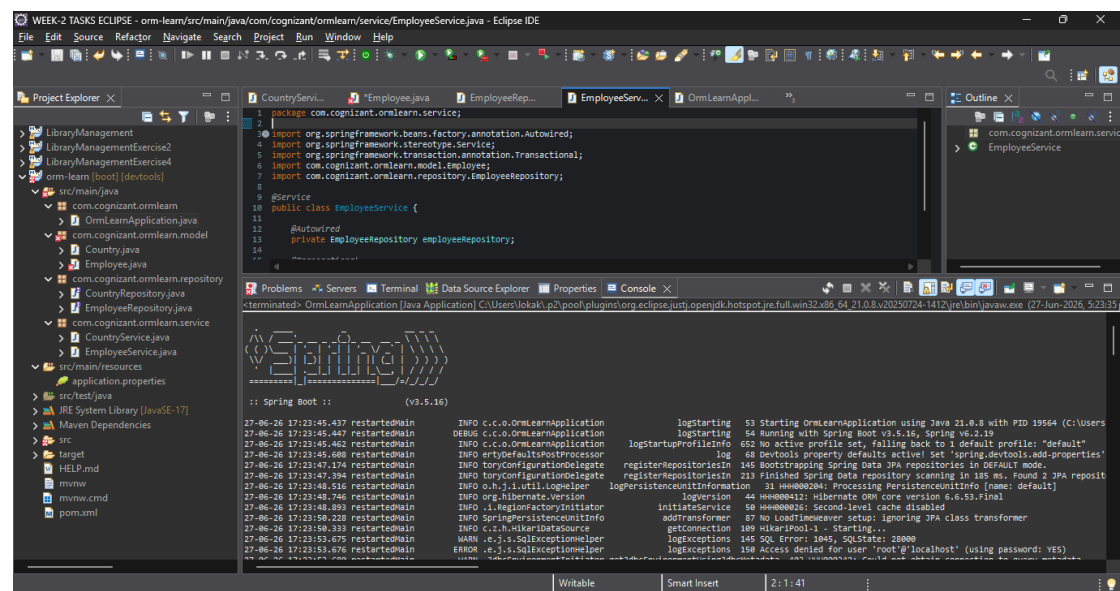
EmployeeRepository.java:

```
CountryServi... X *Employee.java EmployeeRep... X EmployeeServ... OrmL
1 package com.cognizant.ormlearn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5 import com.cognizant.ormlearn.model.Employee;
6
7 @Repository
8 public interface EmployeeRepository extends JpaRepository<Employee, Integer> {
9 }
```

EmployeeService.java:

```
CountryServi... *Employee.java EmployeeRep... EmployeeServ... X OrmLearnAppl... >>_3
1 package com.cognizant.ormlearn.service;
2
3 import org.springframework.beans.factory.annotation.Autowired;
4 import org.springframework.stereotype.Service;
5 import org.springframework.transaction.annotation.Transactional;
6 import com.cognizant.ormlearn.model.Employee;
7 import com.cognizant.ormlearn.repository.EmployeeRepository;
8
9 @Service
10 public class EmployeeService {
11
12     @Autowired
13     private EmployeeRepository employeeRepository;
14
15     @Transactional
16     public void addEmployee(Employee employee) {
17         // Spring Data JPA handles opening the session, transaction management, and closing automatically!
18         employeeRepository.save(employee);
19     }
20 }
```

Output:



The screenshot displays the Eclipse IDE interface. The top editor shows the `EmployeeService.java` file, which is identical to the code provided in the previous block. The left sidebar shows the Project Explorer with the project structure. The bottom console window shows the output of the application, including the Spring Boot logo and various log messages.

```
WEEK-2 TASKS ECLIPSE - orm-learn/src/main/java/com/cognizant/ormlearn/service/EmployeeService.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
Library/Management
Library/ManagementExercise2
Library/ManagementExercise4
orm-learn [boot] (devtools)
src/main/java
com.cognizant.ormlearn
OrmlLearnApplication.java
com.cognizant.ormlearn.model
Country.java
Employee.java
com.cognizant.ormlearn.repository
CountryRepository.java
EmployeeRepository.java
com.cognizant.ormlearn.service
CountryService.java
EmployeeService.java
src/main/resources
application.properties
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
HELP.md
mvnw
mvnw.cmd
pom.xml

Problems Servers Terminal Data Source Explorer Properties Console
<terminated> OrmlLearnApplication [Java Application] C:\Users\lokal\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20250724-1412\jre\bin\javaw.exe (27-Jun-2025, 5:23:33)

Spring Boot (v3.5.16)
27-06-26 17:23:45.437 restartedMain INFO c.c.o.OrmlLearnApplication logStarting 53 Starting OrmlLearnApplication using Java 21.0.8 with PID 19564 (C:\Users
27-06-26 17:23:45.447 restartedMain DEBUG c.c.o.OrmlLearnApplication logStarting 54 Running with Spring Boot v3.5.16, Spring v6.2.19
27-06-26 17:23:45.462 restartedMain INFO c.c.o.OrmlLearnApplication logStartupProfileInfo 652 No active profile set, falling back to 1 default profile: 'default'
27-06-26 17:23:45.688 restartedMain INFO org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 68 Devtools property defaults active! Set 'spring.devtools.add-properties'
27-06-26 17:23:47.174 restartedMain INFO org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 145 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
27-06-26 17:23:47.394 restartedMain INFO org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 213 Finished Spring Data repository scanning in 185 ms. Found 2 JPA reposi
27-06-26 17:23:48.516 restartedMain INFO org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 31 Hibernate: processing PersistenceUnitInfo [name: default]
27-06-26 17:23:48.746 restartedMain INFO org.hibernate.Version logVersion 44 HHH000041: Hibernate ORM core version 6.6.53.Final
27-06-26 17:23:48.893 restartedMain INFO org.hibernate.Version logVersion 58 HHH000042: Second-level cache disabled
27-06-26 17:23:50.220 restartedMain INFO org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 87 No Log4j2 configuration found; using default configuration
27-06-26 17:23:50.333 restartedMain INFO org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 109 HikariPool-1 - Starting...
27-06-26 17:23:53.679 restartedMain WARN org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 149 SQL Error: 1045, SQLState: 28000
27-06-26 17:23:53.679 restartedMain ERROR org.springframework.boot.devtools.restart.RestartScope registerRepositoryesIn 159 Access denied for user 'root'@'localhost' (using password: YES)
```